Infinitely many integer solutions of $x^2 + y^2x + z^2y - x - y + 1 = 0$

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Statement

Theorem 1. *The equation*

$$x^2 + y^2x + z^2y - x - y + 1 = 0 \tag{1}$$

has infinitely many solutions in integers x, y, z. Explicitly, for every pair of integers (u, v) with

$$u^2 - 7v^2 = 1, \tag{2}$$

the triple

$$x = -4v^2\big(2v^2 + 1 + u\big), \qquad y = 4v^2 + 1, \qquad z = 2v\big(2v^2 - 1\big) \tag{3}$$

is a solution of (1)*. Since* (2) *has infinitely many solutions, and $y = 4v^2 + 1$ takes infinitely many values along them, the solutions* (3) *are infinitely many.*

The first members of the family are

n	(u_n, v_n)	x	y	z
0	$(1, 0)$	0	1	0
1	$(8, 3)$	-972	37	102
2	$(127, 48)$	$-43\,646\,976$	9 217	442 272
3	$(2024, 765)$	$-2\,744\,646\,727\,500$	2 340 901	1 790 786 970
4	$(32257, 12192)$	$-176\,781\,544\,692\,719\,616$	594 579 457	7 249 112 703 168

Each line has been checked by direct substitution.

1 Proof

1.1 The polynomial identity

Everything rests on one identity in the polynomial ring $\mathbb{Z}[u, v]$.

Proposition 2. *With x, y, z as in* (3)*,*

$$x^2 + y^2x + z^2y - x - y + 1 \;=\; 16\,v^4\,\big(u^2 - 7v^2 - 1\big). \tag{4}$$

Proof. Put $s = 4v^2$, so that

$$y = s + 1, \qquad z = v(s-2), \qquad x = -\tfrac{s}{2}\,(s + 2 + 2u).$$

Group the left side of (4) as a quadratic in x:

$$x^2 + y^2x + z^2y - x - y + 1 \;=\; x\,\big(x + y^2 - 1\big) + \big(yz^2 - y + 1\big). \tag{5}$$

First factor. Since $y^2 - 1 = s^2 + 2s = s(s+2)$,

$$x + y^2 - 1 = -\tfrac{s}{2}(s + 2 + 2u) + s(s+2) = \tfrac{s}{2}\,(s + 2 - 2u),$$

hence

$$x\,(x + y^2 - 1) = -\tfrac{s^2}{4}\,(s + 2 + 2u)(s + 2 - 2u) = -\tfrac{s^2}{4}\Big((s+2)^2 - 4u^2\Big).$$

Second term. Using $v^2 = s/4$,

$$yz^2 - y + 1 = (s+1)\,\tfrac{s}{4}(s-2)^2 - s = \tfrac{s}{4}\Big((s+1)(s-2)^2 - 4\Big) = \tfrac{s}{4}\big(s^3 - 3s^2\big) = \tfrac{s^3}{4}\,(s-3),$$

because $(s+1)(s-2)^2 - 4 = (s+1)(s^2 - 4s + 4) - 4 = s^3 - 3s^2$. Adding the two pieces,

$$(5) = \tfrac{s^2}{4}\Big(s(s-3) - (s+2)^2 + 4u^2\Big) = \tfrac{s^2}{4}\Big(4u^2 - 7s - 4\Big) = \tfrac{s^2}{4}\cdot 4\big(u^2 - 7v^2 - 1\big),$$

where we used $s(s-3) - (s+2)^2 = -7s - 4$ and $7s + 4 = 28v^2 + 4$. Finally $s^2 = 16v^4$, which gives (4). □

Consequently, whenever $u^2 - 7v^2 = 1$, the right side of (4) vanishes and (3) is an integer solution of (1). (The identity is even in u, so $u \mapsto -u$ gives the second solution $x' = -4v^2(2v^2 + 1 - u)$ with the same y, z; it is the Vieta partner $x' = 1 - y^2 - x$ of x.)

1.2 Infinitely many Pell solutions

Lemma 3. *Define integers u_n, v_n by $u_n + v_n\sqrt{7} = (8 + 3\sqrt{7})^n$, $n \geq 0$. Then $u_n^2 - 7v_n^2 = 1$ for all n, and $0 = v_0 < v_1 < v_2 < \cdots$.*

Proof. Conjugating $\sqrt{7} \mapsto -\sqrt{7}$ gives $u_n - v_n\sqrt{7} = (8 - 3\sqrt{7})^n$, so

$$u_n^2 - 7v_n^2 = (u_n + v_n\sqrt{7})(u_n - v_n\sqrt{7}) = \big((8 + 3\sqrt{7})(8 - 3\sqrt{7})\big)^n = (64 - 63)^n = 1.$$

Expanding $(u_n + v_n\sqrt{7})(8 + 3\sqrt{7})$ gives the recursion $u_{n+1} = 8u_n + 21v_n$, $v_{n+1} = 3u_n + 8v_n$. By induction $u_n \geq 1$ and $v_n \geq 0$, so $v_{n+1} \geq 3u_n + 8v_n > v_n$. □

1.3 Conclusion

For each $n \geq 0$ let (x_n, y_n, z_n) be (3) evaluated at (u_n, v_n). By Proposition 2 and Lemma 3 each triple is an integer solution of (1), and $y_n = 4v_n^2 + 1$ is strictly increasing in n, so the triples are pairwise distinct. This proves Theorem 1. □

2 How the family was found

The proof above is a two-line verification; the content is in the choice (3). The following reductions led to it, and they also explain why the solution set is so sparse (122 solutions with $2 \leq y \leq 5 \cdot 10^4$ in a complete search, all with $x < 0$ and $y \equiv 1 \pmod 4$).

1. Fixed y gives a conic. Completing the square in x, (1) is equivalent to

$$W^2 + 4y\,z^2 = N(y), \qquad W := 2x + y^2 - 1, \qquad N(y) := (y^2-1)^2 + 4(y-1) = (y-1)\,g(y), \quad (6)$$

with $g(y) = y^3 + y^2 - y + 3$. For $y > 0$ this is an ellipse (finitely many points for each y); for $y < 0$ it is a Pell-type hyperbola, but a complete generalized-Pell computation shows it has *no* integer points for $-120 \le y < 0$. (In every solution x is even and $y \equiv 1 \pmod 4$: y is odd because $y \mid x^2 - x + 1$, and reducing $yz^2 = -(x^2 + (y^2-1)x + 1 - y)$ modulo 8 with $y^2 \equiv 1$ first excludes odd x, then forces z even and $4 \mid y - 1$.) So infinitely many solutions must come from infinitely many values of y, not from a Pell family on one fibre. (The surface is a cubic surface whose only $\mathbb{Q}$-rational line is the line at infinity $\{y = 0\}$, so $y = \text{const}$ is its only conic bundle over $\mathbb{Q}$.)

2. $y-1$ is a sum of two squares. Reducing (6) modulo $y-1$ gives $W^2 + 4z^2 \equiv 0 \pmod{y-1}$. If a prime $p \equiv 3 \pmod 4$ divides $y-1$, then $p \mid W$ and $p \mid z$ (as -1 is a non-residue), so $p^2 \mid (y-1)g(y)$; since $g(y) \equiv g(1) = 4 \pmod p$, $p^2 \mid y - 1$, and dividing W, z by p repeats the argument with $(y-1)/p^2$. Hence every such p divides $y - 1$ to an even power. Writing $y - 1 = A^2 + B^2$ and factoring in $\mathbb{Z}[i]$, $W + 2iz = (A + iB)(C + iD)$, the equation becomes

$$C^2 + D^2 + (AD + BC)^2 = g(y), \qquad y = 1 + A^2 + B^2, \quad (7)$$

with $W = AC - BD$, $2z = AD + BC$.

3. The subfamily $A = 0$. If $y-1 = B^2$ is itself a square, (7) reads $D^2 + y\,C^2 = g(y)$. Modulo $y-1$ this forces $C^2 \equiv g(1) = 4$, i.e. $C \equiv \pm 2 \pmod{y-1}$. The choice $C = 2$ gives $D^2 = (y-1)^2(y+3)$, which requires $B^2 + 4$ to be a square (only $B = 0$). The choice $C = y - 3$ gives

$$D^2 = g(y) - y(y-3)^2 = 7y^2 - 10y + 3 = (y-1)(7y-3) = B^2\,(7B^2 + 4),$$

so it suffices that $7B^2 + 4 = E^2$ — the Pell equation $E^2 - 7B^2 = 4$. All its solutions have E, B even ($E^2 - 7B^2 \equiv 2 \pmod 8$ if both were odd), so $E = 2u$, $B = 2v$ with $u^2 - 7v^2 = 1$. Unwinding $W = -BD = -B^2E$, $2z = BC = B(B^2 - 2)$ and $x = (W - y^2 + 1)/2$ gives exactly (3).

The data confirm the mechanism: the solutions $(x, y, z) = (-972, 37, 102)$ and $(-43646976, 9217, 442272)$ found by exhaustive search are the $n = 1, 2$ members, and $D = 96$ at $n = 1$ is precisely B at $n = 2$ ($96 = 2 \cdot 48$).

Remark 4. Since $v_n \sim c\,(8 + 3\sqrt{7})^n$, the family contributes only $O(\log Y)$ solutions with $y \le Y$, in line with the sparse numerical picture. The remaining solutions found by search (e.g. $y = 21, 469, 577, 721, \dots$) are not members of this family (for $y = 577$ one has $y - 1 = 24^2$, but $7 \cdot 24^2 + 4$ is not a square); whether they organise into further families is not addressed here.